Sathwik Karnik

Address: Durand Building, 496 Lomita Mall, Stanford, CA 94305 **Contact:** sathwik@stanford.edu

EDUCATION

Stanford University — **Stanford, CA**
Ph.D. in Aeronautics and Astronautics, Advised by Prof. Somil Bansal, GPA: 4.0/4.0 — 2025 - Present

Massachusetts Institute of Technology (MIT) — **Cambridge, MA**
M.Eng. in Electrical Engineering & Computer Science with a Concentration in Artificial Intelligence — 2022 - 2023
S.B. in Electrical Engineering & Computer Science and Mathematics (Double Major) — 2018 - 2022
GPA: 5.0/5.0

CONFERENCE AND JOURNAL PUBLICATIONS

Karnik, S.*, Kim, J.*, Koyejo, S., Lee, J., Bansal, S. (2026). Steering Away from Memorization: Reachability-Constrained Reinforcement Learning for Text-to-Image Diffusion. Conference on Neural Information Processing Systems (NeurIPS).

Karnik, S., Bansal, S. (2025). Preemptive Detection and Steering of LLM Misalignment via Latent Reachability. Transactions on Machine Learning Research (TMLR).

Hong, Z., Kumar, A., **Karnik, S.**, Bhandwaldar, A., Srivastava, A., Pajarinen, J., Laroche, R., Gupta, A., Agrawal, P. (2023). Beyond Uniform Sampling: Offline Reinforcement Learning with Imbalanced Datasets. Conference on Neural Information Processing Systems (NeurIPS).

Drori, I. et al (including **Karnik, S.**) (2023). From Human Days to Machine Seconds: Automatically Answering and Generating Machine Learning Final Exams. ACM SIGKDD Conference on Knowledge Discovery and Data Mining.

Karnik, S., Lozano-Perez, T., Kaelbling, L.P., Goretkin, G. (2022). Fully Persistent Spatial Data Structures for Efficient Queries in Path-Dependent Motion Planning Applications. International Conference on Robotics and Automation (ICRA).

da Silva, A. M., **Karnik, S.**, Castellanos, P. (2022). Evidential Deep Learning for Polarimeter Data Assimilation. 2022 Advancement of POLarimetric Observations (APOLO) Conference; 2022 Aug 9-12; Washington, D.C.

Karnik, S.*, Ma, S.*, Shao, E.* (2018). A Novel Algorithm to Compute the Cutting Sequence of Billiard Trajectories in Equilateral Triangles. 2018 MAA Undergraduate Poster Session, Joint Mathematics Meetings. 2018 Jan 10-13; San Diego, CA.

WORKSHOP PAPERS AND PREPRINTS

Karnik, S.*, Lee, J. JR.*, Gupta, A., Bansal, S. (2026). When Reasoning Helps Action: Monitoring and Steering Chain-of-Thought in Vision-Language-Action Policies. arXiv:2610.00601. Under review.

Wang, H., **Karnik, S.**, Lim, B., Bansal, S. (2025). Using Language Models as Closed-Loop High-Level Planners for Robotics Applications: A Brief Overview and Benchmarks. IROS AIM-Ctrl Workshop.

Karnik, S.*, Hong, Z.*, Abhangi N.*, Lin, Y., Wang, T., Agrawal, P. (2024). Red Teaming Language-Conditioned Robot Models via Vision Language Models. NeurIPS Workshop on Safe Generative AI.

Karnik, S. (2017). On the Classification and Algorithmic Analysis of Carmichael Numbers. arXiv:1702.08066. Presented at the International Science and Engineering Fair (ISEF) 2017.

INDUSTRY EXPERIENCE

Applied Intuition — **Mountain View, CA**
Software Engineer — 07/2023 – 05/2025
- Developed a performant data-oriented simulator in C++ for autonomous systems.

Jump Trading — **Singapore, Singapore and Chicago, IL**
Quantitative Research Intern — 06/2022 – 08/2022
- Developed predictive models for trading strategies on Indian futures and options and US equities.

NASA Goddard Space Flight Center — **Greenbelt, MD (remote)**
Machine Learning Research Intern — 01/2022 – 06/2022
- Implemented deep evidential regression using TensorFlow for improved model error estimation of aerosol optical depth (AOD).

Uber Advanced Technologies Group (ATG) — **Pittsburgh, PA (remote)**
Software Engineering Intern — 06/2020 – 08/2020
- Designed and developed nearest-neighbor data structures and algorithms for spatial indexing in C++.
- Developed optimized spatial indexing via parallel GPU computation using CUDA C++.

Dexai Robotics **Somerville, MA**
Robotics Software Engineering Intern 06/2019 – 08/2019
· Developed the multi-robot communication and collision-free planning features in C++ for the company product.

TEACHING EXPERIENCE

Computational Sensorimotor Learning (6.8200, previously 6.484) at MIT **Spring 2023**
· Teaching Assistant

Introduction to Machine Learning (6.3900, previously 6.036) at MIT **2019 - 2022**
· Teaching Assistant (2021-2022)
· HKN Tutor (2020-2021)
· Laboratory Assistant (Fall 2019)

SELECTED AWARDS

Qualcomm Innovation Fellowship	2026
Stanford Graduate Fellowship in Science & Engineering (3-year fellowship)	2025
Inducted Member of MIT IEEE Eta Kappa Nu (HKN) Society	2021
MIT Quick Undergraduate Research and Innovation Scholar	2020
HackMIT Top Ten Overall	2019
William Lowell Putnam Mathematical Examination Top 500, Rank 251.5	2018
Joint Mathematics Meetings (AMS and MAA) Undergraduate National Poster Session Presenter	2018
Intel International Science and Engineering Fair (ISEF) 4th Place Grand Award in Math	2017
Intel ISEF 1st Place in Math Air Force Research Laboratory Award	2017
25th National Geographic Bee Champion	2013

MENTORSHIP

Mentor in Safe and Intelligent Autonomy Lab **2026 - Present**
Currently mentoring three MS students on different research projects.

SERVICE

Reviewer: ICLR, ICML (Gold Reviewer Award, Top 25% of Reviewers), NeurIPS, ICRA, IEEE RA-L	2025 - Present
Judge: Hack-Nation Global AI Hackathon, in collaboration with the MIT Sloan AI Club	2025

ASSOCIATIONS

Stanford AIAA Board Member	2026 - Present
IEEE Eta Kappa Nu (HKN) Society	2021 - Present
National Geographic Society (awarded lifetime membership as winner of 2013 National Geographic Bee)	2013 - Present

INVITED TALKS

UCSD Safe Autonomous Systems Lab: "Steering Generative AI Models via Latent Reachability." 2026